

PROJECT OVERVIEW

Machine Learning Technique for Body Movement Recognition using Wearable Devices

Developed as a B.Tech Minor Project

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1. Introduction

Human Activity Recognition (HAR) has emerged as a significant area of research in healthcare, fitness monitoring, sports analytics, and smart wearable technologies. With the rapid advancement of sensor technologies and machine learning algorithms, it has become possible to recognize and classify human activities using data collected from wearable devices.

This project presents a Machine Learning-based Body Movement Recognition System designed to identify and classify different human movements using sensor-generated motion data. The system processes real-time movement information, extracts relevant features, and applies machine learning techniques to recognize activities accurately and efficiently.

By utilizing wearable sensor data instead of camera-based monitoring systems, the solution offers a privacy-friendly, portable, and computationally efficient approach for activity recognition.

2. Problem Statement

Traditional activity monitoring systems often rely on manual observation or camera-based tracking methods. While effective in certain applications, these approaches may introduce privacy concerns, require substantial computational resources, involve complex infrastructure, and limit deployment flexibility.

The challenge is to develop an intelligent system capable of accurately identifying human movements using lightweight sensor data while maintaining real-time performance, user privacy, and ease of deployment. Compared to vision-based systems, sensor-based solutions offer a more compact, portable, and cost-effective alternative, making them suitable for wearable devices and continuous monitoring applications.

This project addresses these challenges by integrating sensor-based data acquisition with machine learning techniques for automated body movement recognition. The proposed approach reduces system complexity, minimizes hardware requirements, and enables efficient activity recognition in a simple and user-friendly manner.

3. Objectives

The primary objectives of this project are:

- Recognize and classify human body movements.
- Utilize machine learning techniques for activity prediction.
- Process wearable sensor data efficiently.
- Enable real-time activity monitoring.
- Improve classification accuracy.
- Support healthcare, fitness applications and defence applications.
- Develop a scalable Human Activity Recognition framework.

4. Methodology

The project follows a structured workflow for activity recognition.

- **Acquisition and Cloud Integration**

The wearable sensing unit, based on the ESP32 microcontroller, continuously collected motion data from the connected sensors. The acquired sensor readings were transmitted to the Arduino IoT Cloud, where the data was stored and managed in real time.

A dedicated data extraction program was developed to retrieve the stored sensor information from the cloud platform. The extracted data was then converted into JSON format, which served as the primary input for the machine learning pipeline. This structured dataset was subsequently used for pre-processing, feature extraction, and activity classification.

- **Data Pre-processing**

The collected raw data undergoes pre-processing to remove noise and improve data quality before analysis.

- **Feature Extraction**

Relevant movement features are extracted from the processed sensor signals to represent various activity patterns.

- **Machine Learning-Based Classification**

The extracted features are supplied to machine learning models that classify different body movements based on learned patterns.

- **Activity Recognition**

The system identifies and categorizes activities in real time, providing meaningful output for monitoring and analysis.

5. System Architecture

The overall architecture consists of multiple stages that transform raw sensor data into meaningful activity classifications.

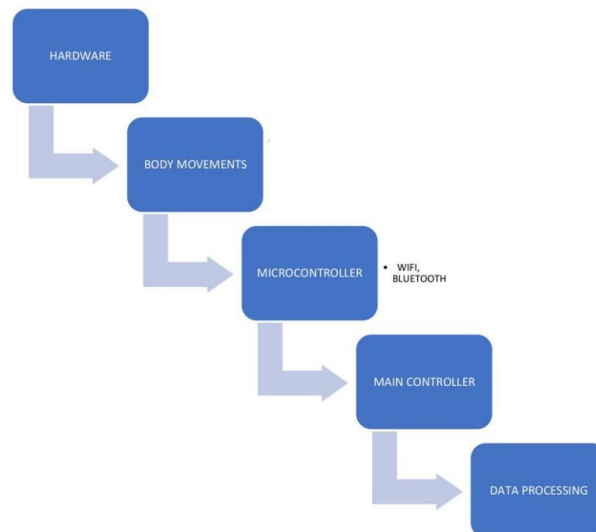


Fig 1: System Architecture

6. Hardware Overview

The project utilizes wearable sensing devices and embedded processing systems for motion data collection.

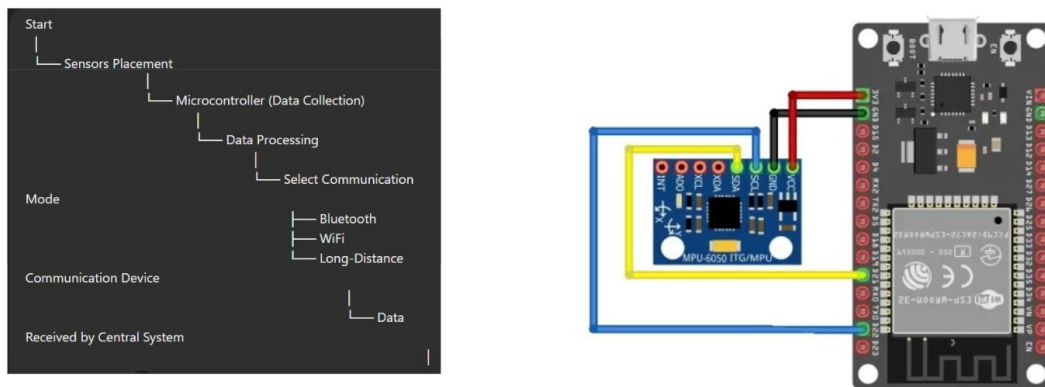


Fig 2: Hardware Connection

Major Components

Accelerometer

Measures linear acceleration along different axes and provides information about body movement patterns.

Gyroscope

Measures angular velocity and orientation changes during movement.

Processing Unit

Handles data collection, pre-processing, and communication with the machine learning environment.

Power System

Provides stable operation for continuous monitoring and data acquisition.

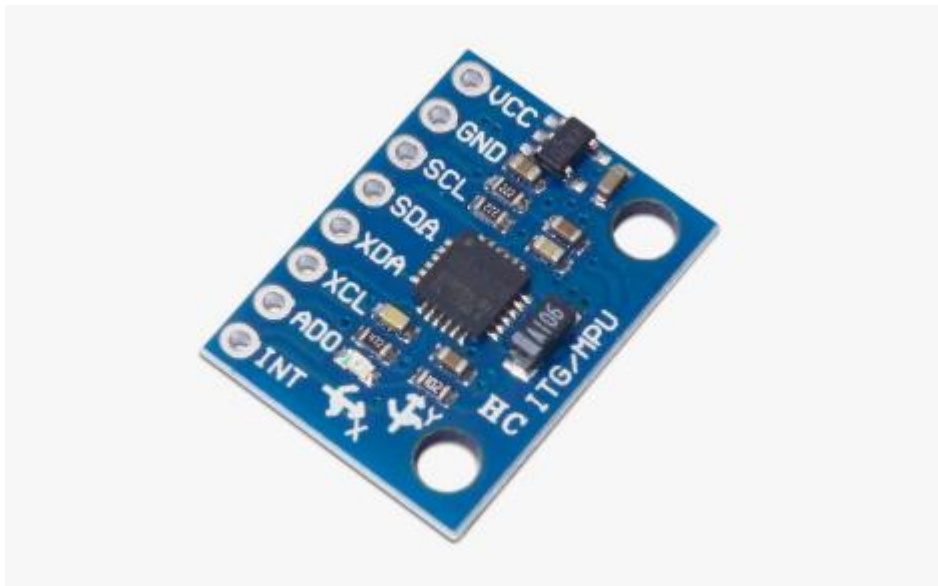


Fig 3: MPU6050 Gyroscope and Accelerometer sensor module.

7. Software Overview

The software platform was developed using machine learning and data analysis tools.

➤ Software Components:

Python

NumPy

Pandas

Scikit-Learn

Data Pre-processing Libraries

Machine Learning Models

➤ Functionalities

Data collection

Data pre-processing

Feature extraction

Activity classification

Performance evaluation

8. Applications

The proposed system has potential applications in multiple domains.

➤ **Healthcare**

Patient monitoring

Rehabilitation tracking

Fall detection systems

➤ **Fitness and Sports**

Exercise monitoring

Performance tracking

Activity analysis

➤ **Human-Computer Interaction**

Gesture recognition

Smart wearable devices

➤ **Smart Living**

Assisted living systems

Activity-aware environments

9. Results

The developed system successfully demonstrated the feasibility of recognizing human body movements using wearable sensor data and machine learning techniques.

Sensor data collected through the ESP32-based sensing unit was transmitted to the Arduino IoT Cloud, where it was stored and managed for further processing. The collected data was subsequently extracted and converted into JSON format, forming the dataset used for machine learning analysis.

To leverage cloud computing resources and accelerate model development, machine learning training was performed using Google Colab. The trained model was developed to classify four fundamental human activities:

Walking
Standing
Sitting
Sleeping

Performance evaluation of the model resulted in an average classification accuracy of approximately 72%, demonstrating the capability of the system to identify different body movement patterns using sensor-based inputs.

For testing and demonstration purposes, the trained model was deployed through a Gradio-based user interface, enabling real-time interaction and activity prediction. A publicly accessible Gradio deployment link with a validity period of 72 hours was used to validate system functionality and user accessibility.

Key Outcomes

Successful collection of real-time sensor data using ESP32 and Arduino IoT Cloud. Fig 4: ML Training Results
Automated dataset generation through JSON-based data extraction.
Cloud-based machine learning training using Google Colab.
Recognition of four human activities: walking, standing, sitting, and sleeping.
Average classification accuracy of approximately 72%.
Successful deployment and testing using a Gradio web interface.
Demonstration of a scalable and portable Human Activity Recognition (HAR) system.

Classification Report:				
	precision	recall	f1-score	
walking	0.76	0.78	0.77	
standing	0.68	0.65	0.67	
sitting	0.72	0.74	0.73	
sleeping	0.70	0.69	0.70	
accuracy			0.72	
macro avg	0.72	0.72	0.72	
weighted avg	0.72	0.72	0.72	

Fig 4: Machine Learning

The results validate the effectiveness of combining wearable sensing technologies, cloud platforms, and machine learning techniques for real-time body movement recognition applications.

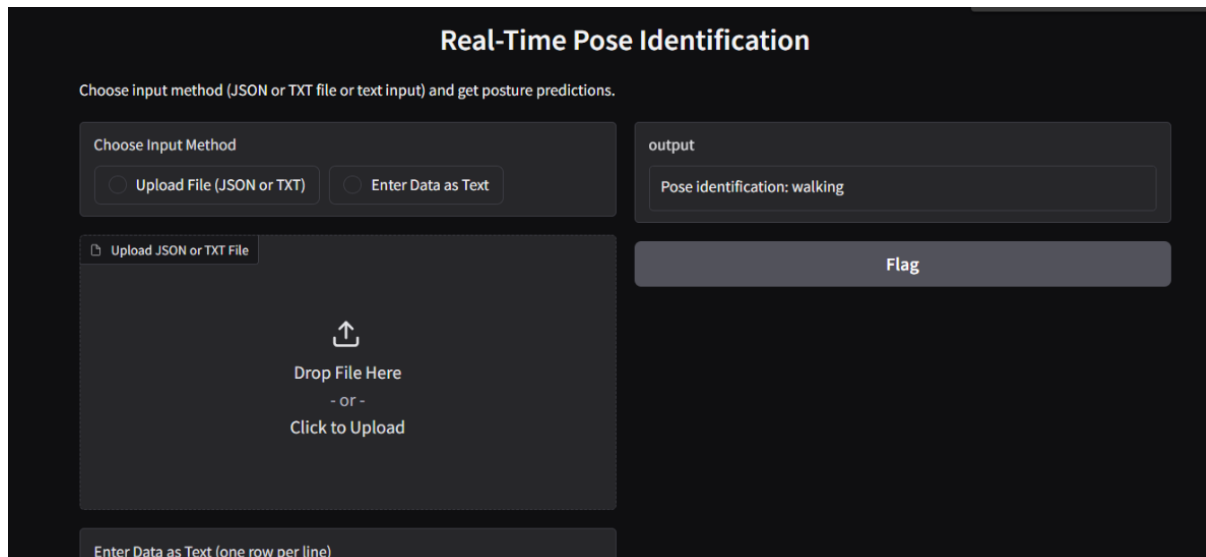


Fig 5: Gradio UI

10. Future Scope

Future enhancements may include:

- Deep Learning-based activity recognition
- Smartwatch integration
- Cloud-based monitoring systems
- Multi-sensor fusion techniques
- Remote healthcare monitoring
- AI-assisted rehabilitation systems
- Advanced sports analytics

These developments can further improve recognition accuracy, scalability, and real-world deployment capabilities.

Conclusion

This project demonstrates the successful implementation of a Machine Learning-based Body Movement Recognition System using wearable sensor data. By combining sensor technologies with intelligent data processing techniques, the system provides an effective solution for Human Activity Recognition.

The project establishes a strong foundation for future developments in healthcare monitoring, fitness tracking, smart wearable devices, and intelligent activity recognition systems.